

Question 1: Answer all the questions.

(8 Marks)

The behavior of charge through a circuit element is shown in **Figure 1a**. The variation of voltage in this element is shown in **Figure 1b**. [4+4]
CO1

- Plot the current vs time graph over the interval $0 \leq t \leq 8s$.
- Plot the power vs time graph for the same interval $0 \leq t \leq 8s$.

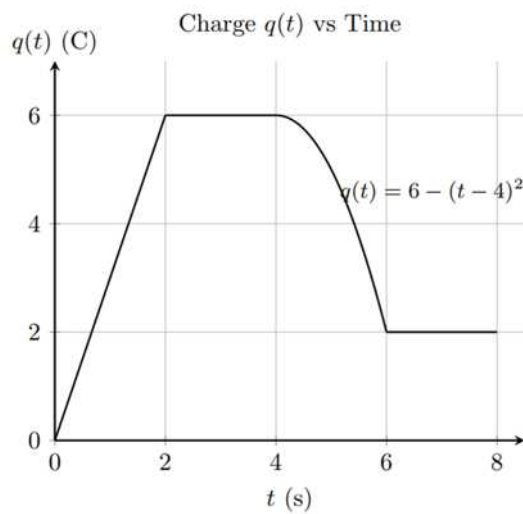


Figure 1a

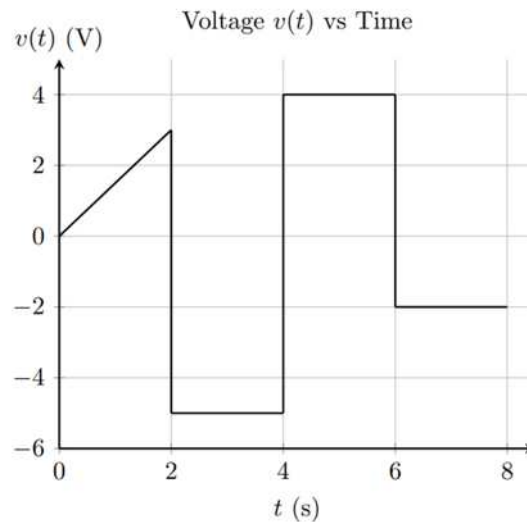


Figure 1b

Question 2: Answer all the questions.

(6 Marks)

Observe **Figures 2a and 2b** and answer the following using KVL, KCL: [3+3]

- For Figure 2a, find the value of the dependent voltage source using KVL. CO2
- Find the value of V_o from Figure 2b

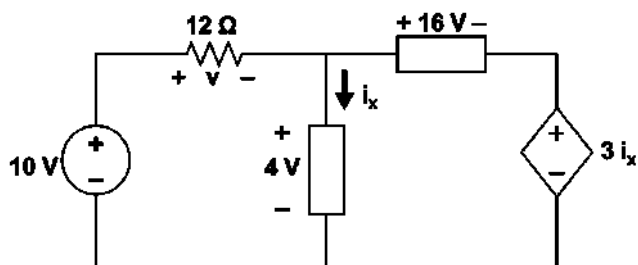


Figure 2a

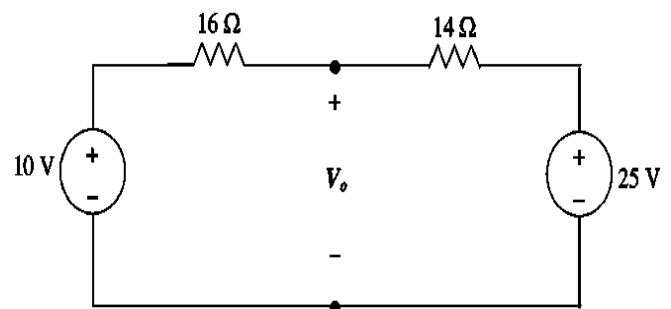
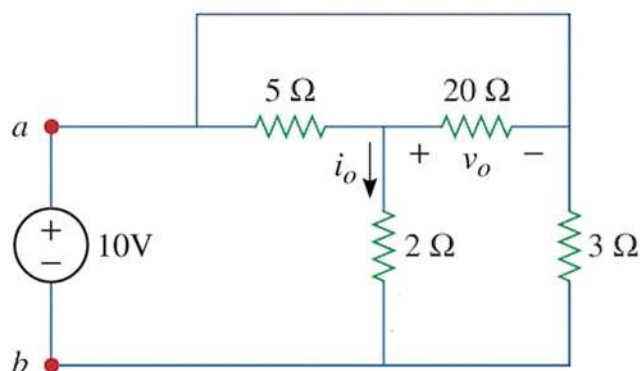


Figure 2b

Question 3: Answer all the questions**(8 Marks)****Determine** the following for the circuit shown in **Figure 3**:

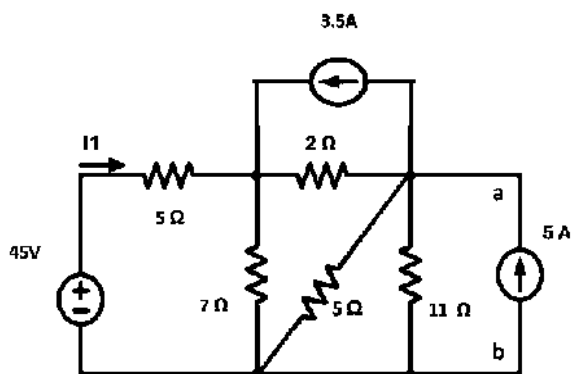
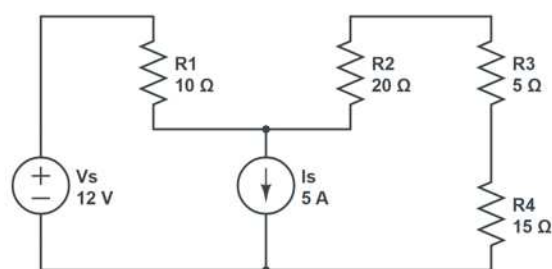
[4+4]

- The Equivalent Resistance seen at the a-b terminals.
- Current i_o and v_o using CDR/VDR.

CO1**Figure 3****Question 4: Answer all the questions.****(8 Marks)**Answer the following questions for the circuit shown in **Figure 4 (a-b)**:

[4+4]

- Using mesh analysis in **Figure 4a**, find the value of I_1 and V_{ab} across the current source.
- Determine** the voltage across the current source using nodal analysis for the circuit shown in **Figure 4b**.

CO2**Figure 4a****Figure 4b**